DRAFT STANDARD CHM-8192-2026 · RESEARCH CONCEPT

CHIMERA II OS

Technical Specification & Reference Document

A dual-stack, 8192-bit research operating system concept covering the IEEE-style draft specification, the R8192/C8192 instruction set, the microkernel and IPv6 routing model, and the zero-copy networking stack.

Generated 9/3/2026

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1. Specification Clauses (1-20)

#	Title	Description
1	**Scope**	This standard defines the architecture, interfaces, and operational requirements for Chimera II OS, a cross-architecture operating system supporting ARM Cortex-M, x86-64, and Chimera R8192/C8192 ISA.
2	**Purpose**	The purpose of this standard is to ensure interoperability, reliability, and performance across heterogeneous hardware platforms, with emphasis on dual-stack IPv6/IPv4 networking and ultra-wide ISA acceleration.
3	**Normative References**	RFC 4291, RFC 6724, RFC 4477, RFC 8421, POSIX.1-2017, Microsoft Winsock2 API.
4	**Definitions**	Defines dual-stack socket, zero-copy buffer, 8192-bit register, tensor instruction, crypto instruction, and kernel routing domain.
5	**System Architecture**	Chimera II OS consists of: bootloader subsystem, microkernel, memory manager, networking subsystem, ISA execution engine, virtualization layer, and user-space runtime.
6	**Bootloader Requirements**	ARM Cortex-M: Thumb-2 startup, vector table, flash loader. x86-64 BIOS: MBR loader, real-mode initialization. UEFI: PEI loader, secure boot, kernel handoff.
7	**Kernel Requirements**	Preemptive scheduler, zero-copy I/O, dual-stack networking, 8192-bit ISA execution engine, memory protection and paging.
8	**Memory Model**	Defines 4 KiB / 2 MiB / 1 GiB pages, MMIO regions, coherent shared memory, and a hypervector accelerator region.
9	**ISA Specification**	Chimera R8192/C8192 ISA: 1024 GPRs (8192-bit), 256 FPRs (8192-bit), 64 tensor registers, 128 parallel lanes, 4096-bit address bus.
10	**Instruction Formats**	Defines a RISC 64-bit fixed format, a CISC variable format (64–4096 bits), and dedicated tensor, crypto, and network instructions.
11	**Networking Requirements**	IPv6-first dual-stack, IPv4-mapped IPv6 addresses, Happy-Eyeballs v2, SLAAC + DHCPv6, DHCPv4 fallback, and a unified routing table.
12	**RFC Compliance**	Chimera II OS must comply with RFC 4291 (IPv6 addressing), RFC 6724 (address selection), RFC 4861 (neighbor discovery), RFC 4862 (SLAAC), and RFC 2131/3315 (DHCPv4/v6).
13	**RFC Bug Corrections**	Chimera II OS must implement a unified DHCP policy engine, consistent DNS merging, route activation delay, and path scoring for ICE.
14	**Socket API Requirements**	A unified API spanning POSIX sockets, Winsock2, and Chimera native sockets.
15	**Zero-Copy Requirements**	The kernel must avoid payload copying, use DMA descriptors, and support scatter-gather I/O.
16	**Routing Requirements**	The kernel must implement an IPv6 routing table, an IPv4 routing table, dual-stack route merging, and per-interface metrics.
17	**Security Requirements**	SHA-3 boot integrity, RSA-8192 crypto acceleration, and memory capability checks.
18	**Virtualization Requirements**	Support for x86-64 virtualization, ARM virtualization, and Chimera ISA emulation.
19	**Performance Requirements**	IPv6 throughput "e 10 Gbps, IPv4 throughput "e 10 Gbps, tensor ops "e 20× speedup.
20	**Conformance**	A system conforms to this standard if all clauses 1–19 are implemented, all referenced RFC bugs are corrected, all dual-stack behaviors match specification, and the ISA simulator passes the validation suite.

2. RFC Dual-Stack Bug Corrections

RFC 4477 — DHCP Dual-Stack Bugs

Bug	Description	Chimera Fix
Conflicting DNS	DHCPv4 and DHCPv6 provide different DNS servers	Unified DHCP policy engine merges DNS lists
Race conditions	IPv4 and IPv6 routes appear at different times	Kernel delays route activation until both families are ready
Split admin domains	IPv4/IPv6 managed separately	Single configuration authority in the kernel

RFC 6724 — Address Selection Bugs

Bug	Description	Chimera Fix
Wrong source address	OS picks a suboptimal IPv6 source address	Kernel implements the full RFC 6724 priority table
Slow IPv4 fallback	IPv6 attempts block IPv4 for too long	Happy-Eyeballs v2 integrated directly in the kernel

RFC 8421 — ICE Dual-Stack Bugs

Bug	Description	Chimera Fix
Broken IPv6 path delays ICE	ICE tries IPv6 first even when the path is broken	Kernel path scoring with early fallback
Multihomed confusion	Multiple interfaces cause the wrong candidate to be selected	Kernel exposes per-interface path metrics

3. ISA Register File & Opcode Table

The state of a Chimera core is a flat structure of ultra-wide registers: 1024 general-purpose registers, 256 floating-point registers, and 64 tensor registers, each 1024 bytes (8192 bits) wide, plus a program counter, stack pointer, and condition flags.

Opcode	Mnemonic	Unit	Description
`0x01`	`ADD`	GPR	8192-bit big-integer addition across the register
`0x02`	`MUL`	GPR	128-lane × 64-bit lane-parallel multiply
`0x20`	`TCONTRACT`	Tensor	Tensor contraction across 128 parallel lanes
`0x21`	`TBIND`	Tensor	Hyperdimensional binding (XOR) of two tensor registers
`0x30`	`MODEXP`	Crypto	RSA-8192 modular exponentiation
`0x31`	`ECCADD`	Crypto	Elliptic curve point addition over 8192-bit coordinates
`0x40`	`SHA3ROUND`	Crypto	SHA-3 / Keccak permutation round
`0x50`	`NETSEND`	Network	Zero-copy transmit to a NIC interface
`0x51`	`NETRECV`	Network	Zero-copy receive from a NIC interface
`0x52`	`CHKSUM8192`	Network	Hardware-accelerated checksum over a buffer

4. Kernel Architecture Overview

- Cooperative round-robin scheduler switching between ready threads (production builds use a preemptive, priority-aware scheduler per Clause 7).

- Unified syscall dispatch for write, read, and networking operations.

- IPv6 routing table with up to 256 routes and 512 neighbor cache entries, matching the RFC 4861 Neighbor Discovery model.

- Longest-prefix-match route lookup, the same algorithm used by real IPv6 forwarding tables.

- Stateless address autoconfiguration (SLAAC, RFC 4862) deriving 128-bit addresses from an EUI-64 interface identifier.

- Happy-Eyeballs v2 (RFC 8305) races an IPv6 attempt against an IPv4-mapped fallback and keeps whichever connects first.

5. Networking Stack Layers

#	Layer	Detail
1	**Socket Layer**	Unified POSIX + Winsock2 API over dual-stack AF_INET6 sockets
2	**Transport**	TCP (RFC 793) and UDP (RFC 768) with zero-copy buffers
3	**Network**	IPv6 (RFC 8200) and IPv4 (RFC 791), routed through a shared table
4	**Neighbor Resolution**	ICMPv6 Neighbor Discovery (RFC 4861) and ARP (RFC 826)
5	**Driver**	NIC abstraction with send/recv hooks and DMA-backed buffers